NuClass: An ontology-driven vision–language foundation model for zero-shot nuclei classification

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Abstract

Nuclei-level classification in histopathology presents persistent challenges due to costly annotations, inter-observer variability, and domain shifts across tissue types and scanners. We introduce ***NuClass****, a vision–language foundation model for zero-shot nuclei classification. NuClass integrates three key components: (1) biologically grounded hierarchical text embeddings constructed from large language models; (2) a nucleus-centered patch design that naturally induces ViT attention to focus on nuclear regions; and (3) large-scale pretraining on over 8.3 million nuclei annotation patches from 11 diverse datasets.*

Across a wide range of benchmarks, NuClass delivers strong and consistent performance, with accuracy gains up to +85% over prior vision–language methods. In addition to quantitative improvements, NuClass provides interpretable and transferable representations aligned with expert-defined semantics, offering a scalable path toward annotation-efficient computational pathology.

1. Introduction

Histopathology remains the gold standard for cancer diagnosis, where the morphological characterization of cell nuclei provides critical information for tumor grading, subtype identification, and treatment planning. However, generating large-scale, high-quality nuclei-level annotations is labor-intensive, prone to inter-observer variability, and often impractical in rare or under-resourced clinical scenarios.

To alleviate the annotation burden, recent efforts have explored vision–language models for weakly supervised and zero-shot learning. One such example is PLIP [14], which adapts the contrastive learning paradigm to digital pathology. PLIP aligns image and text embeddings using a CLIP-style [18] framework and is trained on OpenPath—a large-scale dataset containing over 200,000 histopathology images paired with natural language captions sourced from Twitter, LAION, and other medical domains. While PLIP supports general-purpose retrieval and classification, its supervision is primarily regional/tissue-level. The lack of nucleus-scale granularity limits its ability to distinguish fine-grained cell phenotypes, which are critical for downstream clinical tasks.

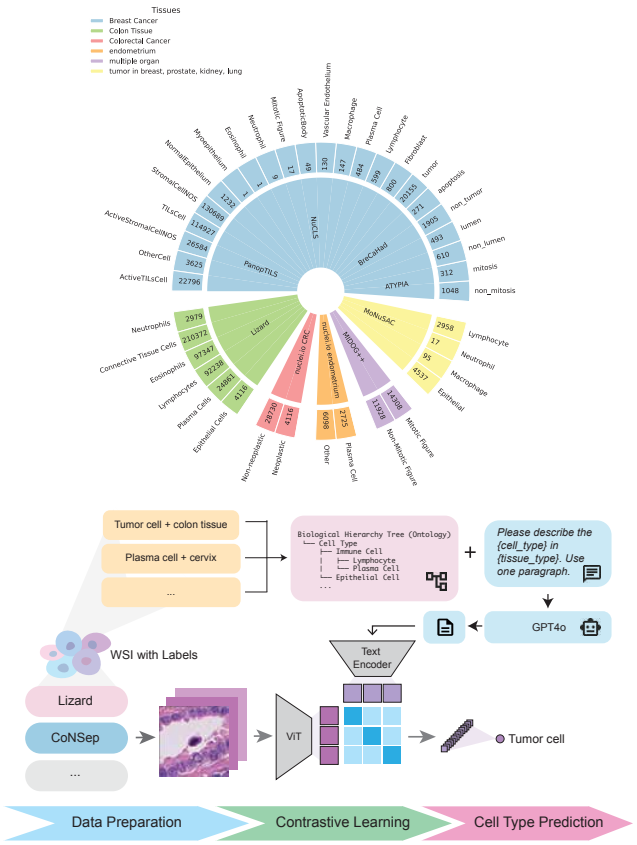


Figure 1. Overview of the training data and NuClass architecture. The framework aligns nuclei-centered image patches with ontology-grounded textual descriptions using contrastive learning, enabling zero-shot cell type classification across diverse tissue and cancer types.

To bridge the granularity gap in nucleus-level classification, we propose **NuClass**, a vision–language founda-

tion model designed for zero-shot cell typing. As illustrated in Figure 1, NuClass is trained on a diverse corpus of histopathology datasets across multiple tissue types and organs. Cell–tissue pairs are structured into a biomedical ontology and used to generate fine-grained, phenotype-aware descriptions via GPT-4o. Corresponding nuclei-centered patches are extracted from whole-slide images and encoded using a vision transformer (ViT). Through contrastive learning, NuClass aligns image and text representations, enabling accurate classification across unseen cell types without per-class supervision. Extensive experiments show that NuClass consistently outperforms prior vision–language models in zero-shot settings and achieves performance comparable to fully supervised baselines, underscoring its potential for real-world applications, such as cell classification and feature representation, both are critical for regional & subject-level analysis.

2. Related Work

Vision–Language Models for Medical Imaging. Domain-adapted vision–language models align histology images with medical text. PLIP fine-tunes CLIP on 208K pathology image–caption pairs, achieving strong zero-shot slide and patch classification. BioViL and BioViL-T [4, 5] extend this to radiology, aligning images with clinical reports, but remain focused on coarse-grained tasks and do not model fine-grained cellular structures.

Zero-Shot Nuclei Classification & Hierarchical Semantics. Zero-shot nuclei classification faces challenges due to annotation scarcity and domain shifts. CLIP underperforms in pathology owing to domain mismatch and limited sensitivity to subtle cellular differences [14]. PLIP addresses pathology-specific alignment but remains coarse-grained. Hierarchy-CLIP [8] and HGCLIP [23] leverage hierarchies for fine-grained tasks, yet lack biomedical context. CPLIP [16] enriches pathology semantics but focuses on slide-level tasks. These limitations motivate NuClass, which integrates biological hierarchies and nucleus-level visual embeddings for fine-grained zero-shot classification.

Nucleus-Level Visual Representation. Current nucleus-level methods primarily rely on segmentation and supervised classification. Hover-Net [9] uses a multi-branch CNN for simultaneous segmentation and classification. CellViT++ [15] leverages pretrained ViT backbones for efficient fine-tuning across datasets but still requires supervision for novel cell types. These constraints inspire zero-shot strategies such as NuClass, which aligns nuclei-focused ViT features with hierarchical textual embeddings.

3. Methodology

NuClass is a dual-stream vision–language framework designed for zero-shot nuclei classification. It aligns visual features from histopathology patches with semantically rich, biologically grounded textual descriptions. Our approach follows a CLIP-style contrastive learning paradigm, extended with hierarchical supervision and ontology-informed sampling to capture fine-grained nuclear phenotypes across tissues and modalities.

3.1. Vision and Text Encoders

Each nucleus-centered patch is resized to 224×224 pixels at 0.25 micron per pixel resolution and processed by a ViT-B/32 backbone [7], following the architecture used in PLIP. We extract the [CLS] token from the final transformer layer and project it into a shared embedding space using a learnable linear head.

For textual supervision, each class description is encoded using a BERT-based text encoder [6]. Mean pooling over the final hidden states, weighted by attention masks, produces fixed-length sentence embeddings. All visual and text features are ℓ_2-normalized before alignment.

3.2. Ontology-Enhanced Text Construction

Each nucleus class (*e.g.*, "Plasma Cell") is described using a two-part textual representation: (1) a rule-based fallback template that captures coarse-to-fine biological hierarchy (*e.g.*, "plasma cell, categorized as immune cell in breast tissue"), and (2) a GPT-4o generated detailed natural-language explanation encompassing morphological, functional, and pathological traits.

This hybrid construction allows NuClass to balance structured biological ontology with linguistic richness, providing a semantically dense supervision signal for contrastive training.

3.3. Contrastive Alignment and Sampling Strategy

We adopt a symmetric InfoNCE loss [21] to align visual and text embeddings. Let $I \in \mathbb{R}^{N\times d}$ and $T \in \mathbb{R}^{N\times d}$ be ℓ_2-normalized image and text representations for a batch of N samples. The cosine similarity matrix $S = IT^\top$ is used to compute the loss:

$$\mathcal{L}_{\text{contrast}} = \frac{1}{N}\sum_{i=1}^{N} -\log \frac{\exp(S_{i,i}/\tau)}{\sum_{j=1}^{N}\exp(S_{i,j}/\tau)} + \frac{1}{N}\sum_{i=1}^{N} -\log \frac{\exp(S_{i,i}/\tau)}{\sum_{j=1}^{N}\exp(S_{j,i}/\tau)}, \quad (1)$$

where τ is a temperature hyperparameter.

To encourage biologically meaningful organization of the embedding space, we apply ontology-aware negative sampling: each batch includes **soft negatives** from similar biological lineages (e.g., immune cell subtypes) and **hard negatives** across distinct lineages (e.g., immune vs. stromal).

3.4. NuClass Variant Definitions

To support diverse downstream tasks, we define three variants of NuClass, each emphasizing different levels of hierarchical structure and linguistic detail:

- **NuClass**$_{\text{Base}}$: employs simple, rule-based templates without hierarchy or natural-language enhancements.
- **NuClass**$_{\text{Hier}}$: extends the Base setting with explicit ontology-driven hierarchical encoding.
- **NuClass**$_{\text{Natural}}$: replaces the templates with detailed LLM-generated natural-language descriptions without explicit hierarchy.

These variants are evaluated across both supervised and zero-shot settings to disentangle the roles of structured and descriptive text supervision.

4. Experiments

We conduct a comprehensive evaluation to assess the performance, generalization, and interpretability of NuClass under multiple experimental settings. Our experiments include: (1) overall performance and an ablation study across different NuClass variants to isolate the effects of hierarchical supervision and natural-language richness; (2) direct comparisons against existing SOTA models, including PLIP and CellViT++; (3) zero-shot transfer evaluations on unseen datasets to measure robustness under distribution shifts; and (4) qualitative analyses based on attention localization and UMAP projections to investigate model interpretability.

4.1. Experimental Setup

All NuClass variants are trained for a total of $4,000$ steps with a batch size of 512, using identical optimization settings. To preserve clinical realism, no artificial class balancing is applied; models are trained under the naturally imbalanced class distributions.

4.2. Comparison with PLIP on Diverse Histopathology Benchmarks

We compare **NuClass**$_{\text{Hier}}$, fine-tuned with hierarchical supervision, to zero-shot PLIP on nine benchmarks (Table 1). NuClass$_{\text{Hier}}$ builds on PLIP's backbone but incorporates ontology-driven prompts and diverse training data.

We use a held-out 10% split for each dataset and report accuracy. Data splitting is conducted on patient-level to avoid potential training data leakage. 95% confidence intervals was calculated via $1,000$ bootstrap resamples of the predictions and the results are presented in Table 1. The last column shows absolute accuracy gains of NuClass$_{\text{Hier}}$ over PLIP.

4.3. Ablation Study: Textual Supervision Strategies

We evaluate the three NuClass variants introduced in Section 3—NuClass$_{\text{Base}}$, NuClass$_{\text{Hier}}$, and NuClass$_{\text{Natural}}$—to isolate the effects of hierarchical structure and natural-language richness in textual supervision. Table 2 summarizes the classification performance across nine benchmark datasets.

Table 1. Validation accuracy (%) of zero-shot PLIP vs. fine-tuned NuClass$_{\text{Hier}}$. 95% CI in parentheses.

Dataset	PLIP (zero-shot)	NuClass$_{\text{Hier}}$ (fine-tuned)	Improvement
CoNSeP [10]	10.32 (±2.59)	65.13 (±2.59)	+54.80
Plasma [12]	43.79 (±5.29)	87.28 (±3.55)	+43.50
NuCLS-Single [2]	23.55 (±1.23)	66.48 (±1.23)	+42.90
CRC_Tumor [13]	89.84 (±0.48)	86.26 (±0.54)	–3.60
Lizard [11]	53.06 (±0.48)	71.02 (±0.44)	+18.00
MIDOG++ [3]	47.04 (±1.42)	80.23 (±1.42)	+33.20
BreCaHad [1]	10.25 (±1.25)	95.19 (±0.88)	+84.90
MoNuSAC [22]	34.47 (±2.75)	89.53 (±1.77)	+55.06
PanopTILS [17]	1.29 (±0.45)	72.79 (±1.45)	+71.50

NuClass$_{\text{Hier}}$ consistently improves classification accuracy over NuClass$_{\text{Base}}$ (Table 2), particularly on datasets with fine-grained morphological distinctions such as Lizard (+13.71%) and PanopTILS (+27.12%). This improvement suggests that hierarchical supervision effectively reduces intra-class ambiguity and enhances fine-grained discrimination under supervised learning.

In contrast, NuClass$_{\text{Natural}}$ yields only marginal gains over NuClass$_{\text{Base}}$ in validation accuracy. However, as demonstrated later in our zero-shot experiments (Section 4.5), the linguistic richness of NuClass$_{\text{Natural}}$ substantially improves generalization to unseen distributions. These results highlight a trade-off: hierarchical encoding strengthens supervised fine-grained classification, whereas natural-language supervision enhances open-world adaptability.

Together, these findings validate our design choice of integrating both structured ontology and descriptive natural language to balance supervised accuracy and zero-shot robustness in NuClass.

Table 2. Ablation study of NuClass model's classification accuracy (%) across nine datasets' testing splits. The best result for each dataset is highlighted in **bold**, and the second-best is underlined.

Dataset	NuClass$_{\text{Base}}$	NuClass$_{\text{Hier}}$	NuClass$_{\text{Natural}}$
Lizard	68.57	**82.28**	<u>72.63</u>
MIDOG++	77.30	**80.52**	<u>78.75</u>
BreCaHad	89.24	**95.69**	<u>93.06</u>
CoNSeP	71.79	**73.97**	<u>73.77</u>
CRC_Tumor	<u>93.82</u>	89.26	**94.11**
NuCLS-Single	55.47	**66.00**	<u>63.75</u>
Plasma	92.85	**95.83**	<u>93.64</u>
MoNuSAC	**92.03**	<u>91.12</u>	89.11
PanopTILS	52.16	**79.28**	<u>62.03</u>

4.4. Comparison with Fully Finetuned CellViT++

We benchmark **NuClass**$_{\text{Hier}}$ against CellViT++ under full supervision, as presented in Table 3. NuClass$_{\text{Hier}}$ matches and surpasses CellViT++ across multiple datasets, with

particularly notable improvements on Plasma dataset and MoNuSAC. These results suggest that hierarchical semantic alignment enhances sample efficiency and generalization, even in fully supervised settings.

Table 3. Validation accuracy (%) of CellViT++ and $\text{NuClass}_{\text{Hier}}$ (fine-tuned).

Dataset	CellViT++	$\text{NuClass}_{\text{Hier}}$	Improvement
Plasma	90.27	**97.07**	+6.80
CRC_Tumor	98.01	**98.26**	+0.30
MoNuSAC	87.14	**92.17**	+5.00

4.5. Case Study: Zero-Shot Generalization across Datasets

We assess zero-shot generalization by testing **NuClass**$_{\text{Natural}}$ without any fine-tuning on three unseen datasets shows in table 4). $\text{NuClass}_{\text{Natural}}$ consistently doubles or nearly triples accuracy compared to PLIP under zero-shot conditions. The strongest gains occur in clinically complex datasets like ATYPIA and MoNuSAC, underscoring the benefit of GPT-driven descriptions for model generalization.

Table 4. Zero-shot performance of PLIP and $\text{NuClass}_{\text{Natural}}$ on three datasets (10% held-out splits).

Dataset	Model	Accuracy	Balanced Acc.	Avg. Precision
MoNuSAC [22]	PLIP	15.88	29.88	**41.45**
	NuClass$_{\text{Natural}}$	**34.47**	**52.84**	40.94
ATYPIA [19]	PLIP	23.89	52.22	86.98
	NuClass$_{\text{Natural}}$	**47.79**	**63.99**	**90.41**
BreaKHis [20]	PLIP	9.60	12.50	13.59
	NuClass$_{\text{Natural}}$	**17.73**	**12.63**	**23.56**

4.6. Qualitative Analysis

Attention Localization. Figure 2 compares ViT attention maps from PLIP and NuClass. Across datasets, NuClass shows sharper focus on nuclei—even with padding from image boundaries—while PLIP spreads attention broadly. This supports the benefit of nucleus-aware representation learning.

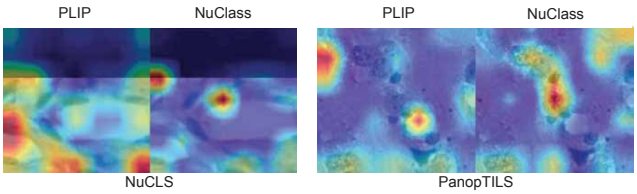
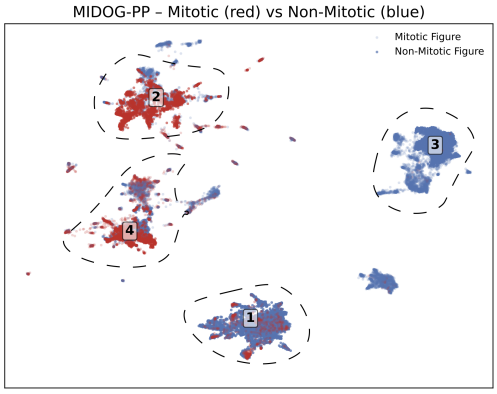
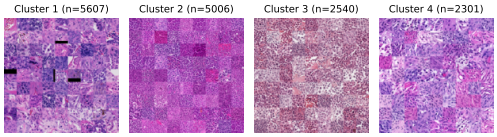


Figure 2. Visualization of ViT attention heatmaps comparing PLIP and NuClass across two datasets: NuCLS and PanopTILS.



(a) UMAP projection of NuClass embeddings on MIDOG++. Color indicates the ground truth.



(b) Patch grids sampled from each cluster. NuClass consistently distinguishes mitotic from non-mitotic nuclei, even under diverse visual appearances.

Figure 3. UMAP-based visualization of NuClass on MIDOG++.

UMAP Projection. Figure 3 illustrates the UMAP projection of NuClass embeddings on the MIDOG++ dataset. Notably, NuClass achieves clear semantic separation between mitotic and non-mitotic nuclei across all clusters, despite intra-class visual heterogeneity. This alignment demonstrates the model's ability to generalize across domains and stain variations—enabled by its hierarchical text-driven supervision.

5. Conclusion

We introduced **NuClass**, an ontology-driven vision–language foundation model for zero-shot nuclei classification. By integrating hierarchical ontologies, natural-language supervision from LLM-generated descriptions, and contrastive learning, NuClass learns semantically aligned representations across diverse cell types and tissue contexts. It achieves strong zero-shot classification performance, approaching that of fully supervised baselines. These results underscore NuClass's potential as a scalable, annotation-efficient approach for real-world pathology AI, paving the way for broader adoption in clinical diagnostics and biomedical research.

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